

Atrial-to-conduction paracrine signaling shapes a spatial maturation gradient in the human fetal heart at PCW12

PantheonOS PCW12 Spatial Analysis Pipeline
Generated report; corresponding workspace:
pantheonos-reproducibility/human-heart

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Abstract

The human cardiac conduction system (CCS) emerges during the first trimester from a heterogeneous pool of myocytes that must coordinate their identity with adjacent atrial and ventricular working myocardium. Whether atrial myocardium (ACM) actively shapes the molecular state of neighbouring conduction cells in the early fetal heart remains poorly resolved. Here we use a 100,000-cell three-dimensional MERFISH atlas of the post-conception week 12 (PCW12) human heart, paired with an entropic optimal-transport imputation against a matched scRNA-seq reference [1], to expand the spatial gene panel from 238 to 5,187 genes and recover a genome-scale spatial cell–cell interactome using Spateo [2]. Re-running the analysis with this expanded panel grew the testable ligand–receptor (LR) catalogue from ~ 11 to 388 pairs and revealed 159 spatially significant interactions spanning neuregulin–ErbB, ECM–integrin, Eph–ephrin and semaphorin modules. Focusing on the conduction compartment, ACM-derived NRG3 and SEMA3A emerge as the dominant inputs to core conduction cells (CCC). A nearest-neighbour spatial test confirmed that NRG3 expression in ACM and ERBB4 expression in adjacent CCC are spatially coupled ($\rho = +0.22$, $p_{\text{perm}} < 10^{-3}$). Unsupervised clustering of the 292 CCC on a curated receptor and CCS-transcription-factor panel resolved seven transcriptional subtypes with clearly interpretable signatures, including an AV-node-like (TBX2/TBX3⁺), SAN-like (SHOX2⁺ NKX2-5⁺) and TBX18⁺ sinus-venosus subset. Across the full CCC pool, ACM-niche scores for NRG3 and SEMA3A were systematically *negatively* correlated with conduction-system transcription factors (NKX2-5, TBX3, TBX5, SHOX2, TBX18; FDR $\leq 10^{-3}$). Together, the data support a model in which proximity to atrial paracrine input sustains a less-differentiated CCC pool while specified pacemaker, AV node and sinus-venosus-derived cells reside further from the source.

Keywords: cardiac conduction system; spatial transcriptomics; MERFISH; optimal transport; cell–cell interaction; neuregulin; ErbB4; SEMA3A; TBX3; SHOX2.

1 Introduction

The cardiac conduction system (CCS) — comprising the sinoatrial node (SAN), the atrioventricular node (AVN), the bundle of His and the Purkinje fibre network — is responsible for orchestrating the sequential contraction of the four chambers of the heart. Its morphogenesis begins early in cardiac development from heterogeneous progenitor pools, and aberrations of this process underlie a substantial fraction of congenital arrhythmias and conduction disorders [3, 4]. Decades of mouse genetics have established a core transcriptional network governing CCS specification and function: *Tbx3* is required for the SAN/AVN gene program [5]; *Shox2* represses *Nkx2-5* in the sinus venosus

to permit pacemaker identity [6, 7]; *Tbx18* defines the head of the SAN and venous pole [8, 9]; and the funny current generated by HCN channels confers spontaneous diastolic depolarisation to pacemaker cells [10].

In parallel, ligand–receptor signalling has been shown to shape CCS differentiation. Neuregulin–ErbB signalling is essential for cardiac development [11, 12]; conditional loss-of-function studies further established that NRG-1 promotes formation of the murine cardiac conduction system [13], and SEMA3A is required for normal cardiac rhythm by patterning sympathetic innervation [14]. However, almost all of this mechanistic knowledge derives from mouse models or in-vitro systems. The corresponding paracrine architecture in the early human fetal heart — where the conducting compartment is intimately interleaved with working myocardium and contains multiple molecularly distinct subtypes — has been difficult to resolve because the available spatial technologies sacrifice gene breadth for spatial resolution.

Recent advances in image-based spatial transcriptomics [15, 16] and 3D spatiotemporal modelling [2] now enable cell-resolution interrogation of intact human organs, but at the cost of restricted gene panels (typically $\lesssim 300$ genes) that exclude most ligand–receptor pairs. Optimal transport (OT) provides a principled remedy: by aligning a high-content scRNA-seq reference to a spatial sample in an entropically regularised coupling, OT permits gene imputation onto unmeasured genes [1, 17].

In this report we apply this strategy to a 100,000-cell 3D MERFISH atlas of the human heart at PCW12, paired with a stage-matched scRNA-seq reference. Imputing $\sim 5,000$ highly variable genes plus a curated heart-disease gene panel multiplies the testable LR landscape by ~ 35 -fold and reveals dominant atrial-to-conduction paracrine axes that are absent from the panel-restricted analysis. We then validate the strongest axis (NRG3→ERBB4) with a spatial nearest-neighbour test, dissect heterogeneity within the core conduction compartment, and demonstrate that proximity to ACM-derived ligands is anti-correlated with CCS transcription factor expression — pointing to a niche-driven maturation gradient.

2 Results

2.1 Expanded MOSCOT imputation enables genome-scale spatial CCI

The MERFISH panel measures 238 genes across 100,000 cells in 3D, of which only ~ 60 overlap the Spateo human ligand–receptor database; direct application of cell–cell interaction analysis yielded merely ~ 11 testable LR pairs and a single biological axis (DLL4–NOTCH2). To overcome this, we used the entropic OT framework MOSCOT [1] to map a stage-matched PCW12 scRNA-seq atlas ($32,870 \text{ cells} \times 31,019 \text{ genes}$) onto the spatial sample (Methods). The mapping was performed in five batches of 2×10^4 spatial cells against the full SC reference, with local-PCA-based costs and asymmetric marginals ($\tau_a=1, \tau_b=0.9$). Imputation used the union of the top 5,000 highly variable genes (HVGs) and a curated panel of 247 heart disease genes, intersected with the SC reference, yielding 5,187 imputed genes per spatial cell.

Coverage of the Spateo LR database increased from 2 ligands and 3 receptors (panel-only) to 185 and 189 respectively, expanding the testable ordered LR catalogue from ~ 11 to 388 (Table 1). Re-running the Spateo `find_cci_two_group` analysis on a 20,000-cell stratified subsample across the eight largest cell types (ACM, VCM, FB, Endocardial, MuralCells, NC, CoreConductionCells, TzConductionCells) returned non-empty results for all 56 ordered cell type pairs and flagged 159

2.2 Atrial myocardium signals to core conduction cells via NRG3 and SEMA3A

Focusing on the conducting compartment, we ran `Spatio find_cci_two_group` on six ordered pairs centred on `CoreConductionCells` (CCC) and its immediate neighbours `ACM`, `NC` (neural crest) and `TzConductionCells`, with $N_{\text{perm}}=50$. Across the `ACM`→`CCC` pair (the most populated of the six), 18 LR pairs reached significance. The two strongest signals were **NRG3–ERBB4** ($n=79$ co-expressing cell pairs, ratio 0.44) and **SEMA3A–PLXNA4** ($n=71$, ratio 0.40) (Figure 2a). The NRG3–ERBB4 axis is the spatial-omics correlate of an interaction with deep developmental support: NRG-1 has been shown to promote murine CCS formation in vivo [13] and the wider NRG/ErbB axis is required for cardiogenesis in mice [11, 12]. SEMA3A is required for normal cardiac rhythm in vivo [14]; its appearance here as the second-strongest `ACM`→`CCC` signal at PCW12 is consistent with a guidance / repulsion role in shaping the AV junction.

The reverse direction (`CCC`→`ACM`) yielded only six significant pairs, dominated by `EFNB2–EPHA4`, `FGF10–FGFR2` and `BMP4–BMP2`; signalling is therefore strongly asymmetric, with `ACM` as the dominant sender. An analogous analysis of the `NC`→`CCC` pair recovered the canonical neural-crest→conduction-system signal `SEMA3C–PLXNA4` (ratio 0.75) and a bidirectional `COL1/COL4–integrin ECM` axis (Figure 2b). The `TzConductionCells–CCC` pair was largely silent, likely reflecting both the small population of transitional cells in the subsample and limited contact between the two compartments at this developmental stage.

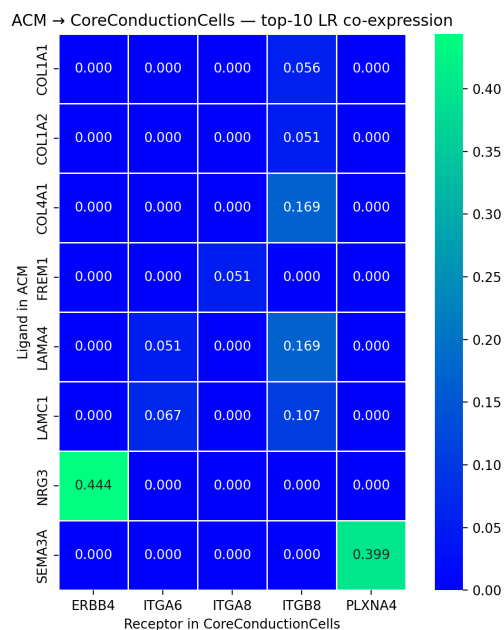
2.3 NRG3→ERBB4 paracrine coupling is spatially coherent

A high LR co-expression count is necessary but not sufficient evidence for paracrine coupling: ligand and receptor could be expressed at the right ratios in the right cell types yet not be *spatially adjacent*. To test for spatial coherence, we returned to the full imputed atlas (`CCC` $n=292$, `ACM` $n=9,955$) and, for each `CCC` cell, recorded the ligand expression (NRG3 or SEMA3A) of its single nearest `ACM` neighbour in 3D. We then computed Spearman correlation against the `CCC` cell’s receptor (ERBB4 or PLXNA4) and assessed significance against a 1,000-permutation null in which `CCC–ACM` pairings were randomly re-shuffled (Methods).

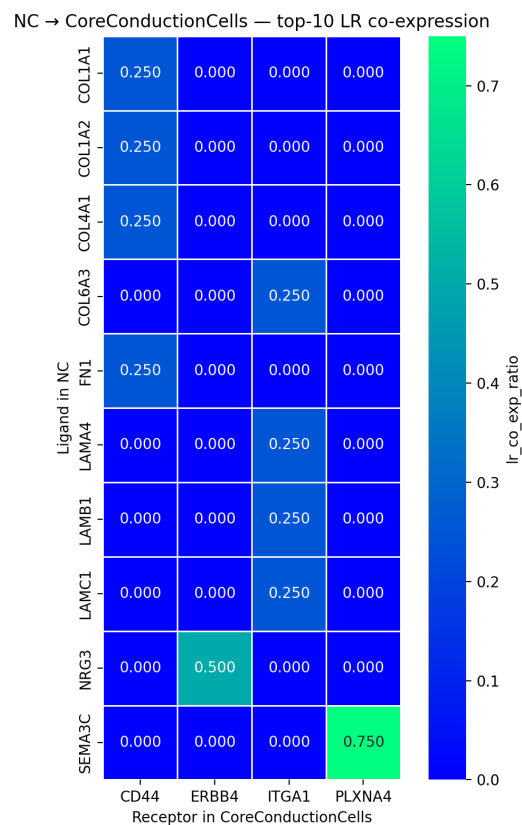
The NRG3→ERBB4 axis showed a clean positive correlation ($\rho = +0.219$, $p_{\text{perm}} < 10^{-3}$) that fell well outside the permutation null (range ± 0.13 ; Figure 3b). The SEMA3A→PLXNA4 axis was significantly *negative* ($\rho = -0.121$, $p_{\text{perm}} = 0.037$): PLXNA4-high `CCC` sit *further* from SEMA3A-high `ACM`, consistent with the known repulsive guidance function of class-3 semaphorins. Three-dimensional visualisation of `ACM` and `CCC` populations confirmed that ERBB4-high `CCC` preferentially occupy regions adjacent to NRG3-high `ACM`, while PLXNA4 expression in `CCC` was spatially less structured (Figure 3a,c).

2.4 Core conduction cells partition into seven transcriptional subtypes recovering canonical CCS biology

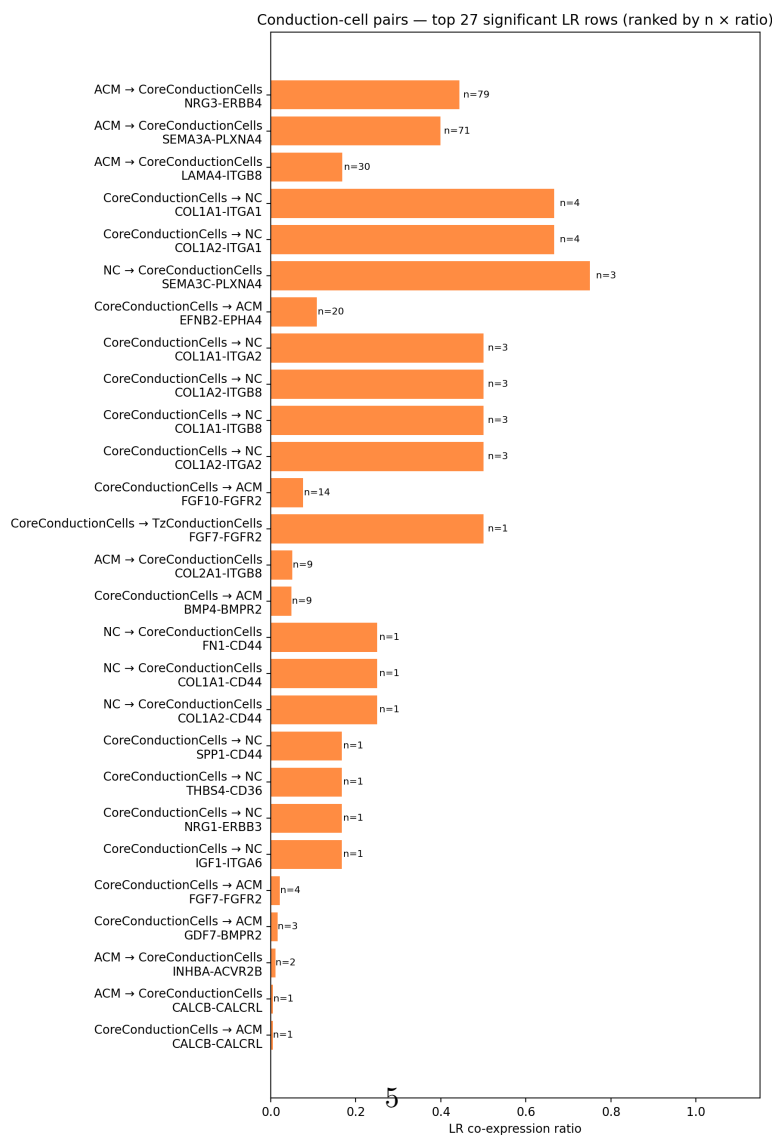
A single `CoreConductionCells` label on the spatial atlas masks substantial molecular heterogeneity. To dissect this, we extracted the full set of 292 `CCC` cells from the imputed atlas and re-clustered them on a curated panel of 8 receptors (ERBB3, ERBB4, FGFR2, PLXNA4, EPHA4, ITGA1, ITGB8, BMP2) and 10 CCS transcription factors (TBX3, TBX5, TBX18, TBX2, NKX2-5, ISL1, SHOX2, HCN1, GJA5, GJA1). Note that HCN4 and CNTN2 were not retained in the 5,187-HVG



(a) ACM → CCC



(b) NC → CCC



(c) Conduction-pair overview: significant LR pairs across all six CCC-

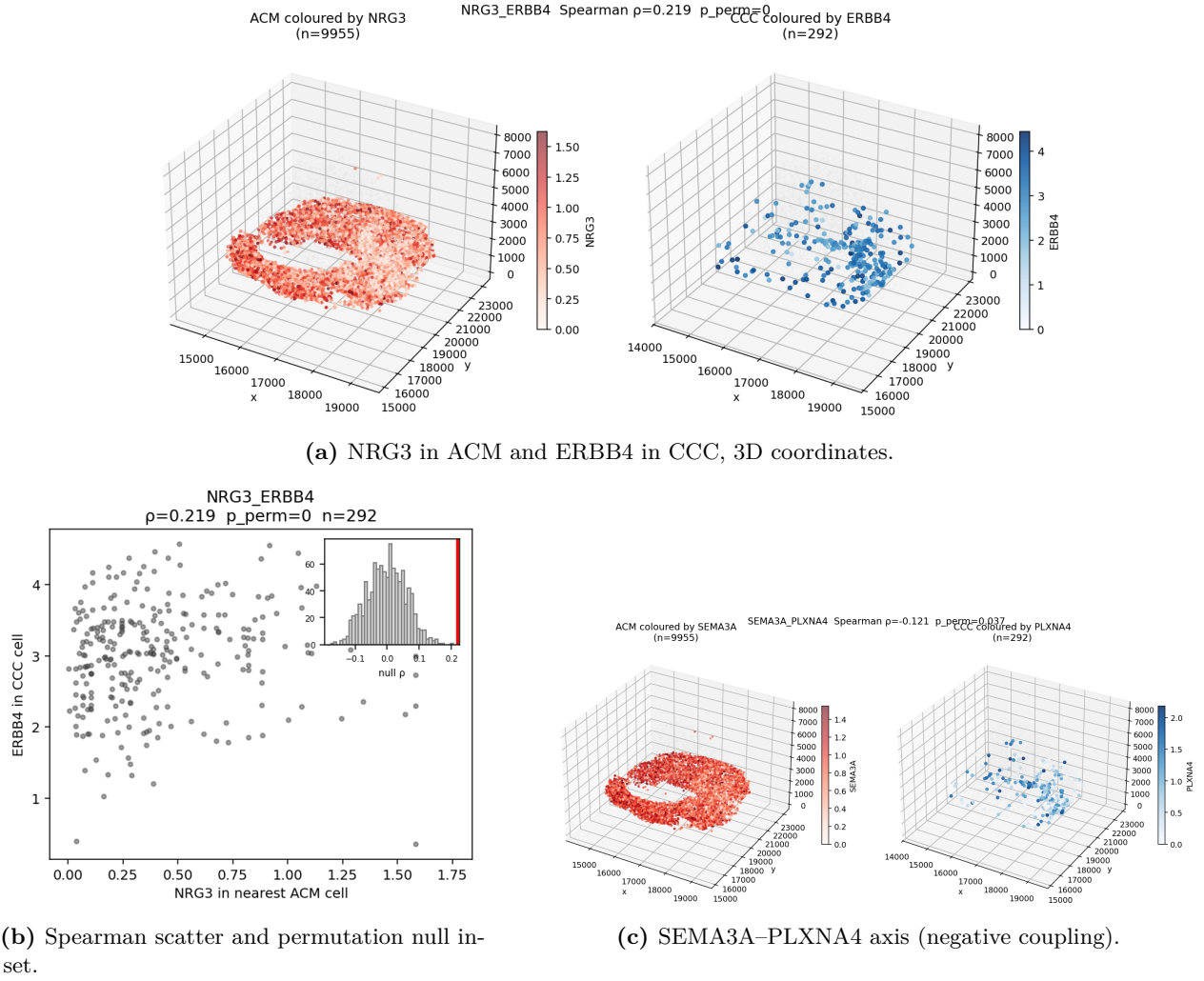


Figure 3. Spatial validation of paracrine coupling. The NRG3→ERBB4 axis is positively coupled in space; the SEMA3A→PLXNA4 axis is significantly anti-coupled, consistent with semaphorin’s repulsive role.

imputed set; HCN1 was used as the closest available pacemaker-channel surrogate (also expressed in His–Purkinje tissue). After Scanpy preprocessing (PCA, neighbour graph) [18], Leiden community detection [19] returned seven subclusters (Figure 4a) with clearly interpretable signatures (Figure 4c, Table 2).

Table 2. CCC subclusters and their inferred CCS-subtype identity.

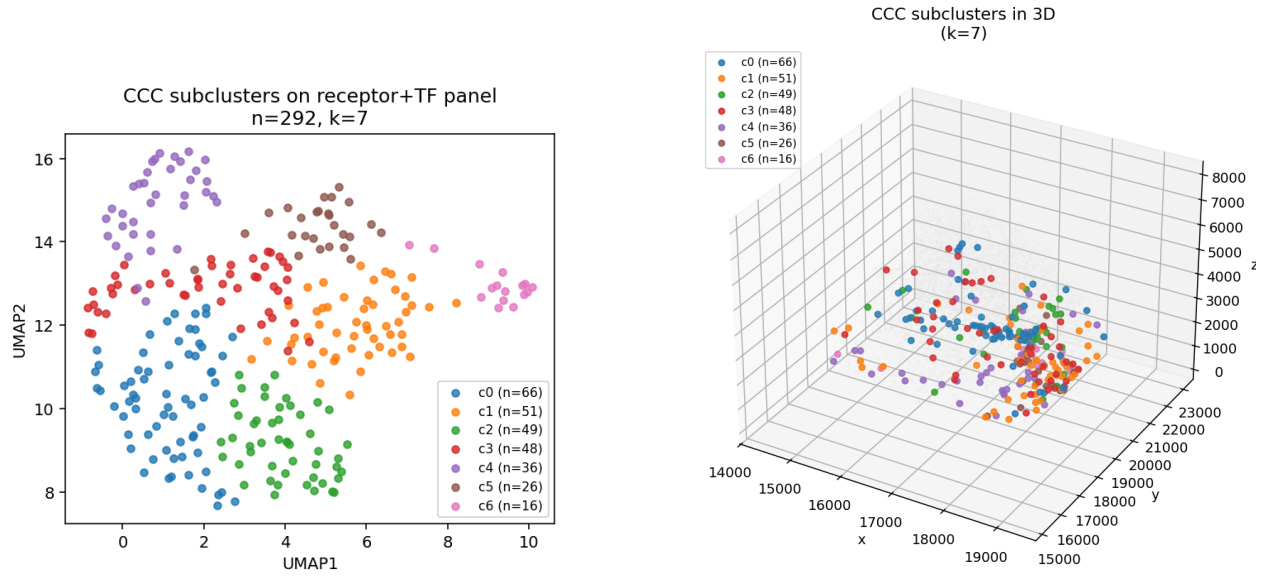
Cluster	<i>n</i>	Signature	Inferred identity / supporting refs
c1	51	HCN1↑ GJA5↑ FGFR2↑	Pacemaker / conducting (canonical HCN ⁺ [10]; GJA5 ⁺)
c2	49	TBX2↑ TBX3↑ EPHA4↑	AV-node-like (TBX3 imposes pacemaker program [5])
c6	16	SHOX2↑ NKX2-5↑ BMPR2↑	SAN-like (SHOX2 is the SAN master regulator [6, 7])
c5	26	TBX18↑ ALDH1A2↑	Sinus-venosus / pro-epicardial origin [8, 9]
c3	48	ISL1↑ GJA1↑	Transitional / atrial–CCS border
c0	66	ERBB3↑ ITGA1↑ ALCAM↑	ECM / vascular-niche subset
c4	36	low everything; MRC2↑ FKTN↑	Poorly-differentiated / fibrotic-niche

The five biologically labelled clusters (c1, c2, c3, c5, c6) account for ~65% of the CCC pool. The recovery of an AV-node-like (TBX2⁺/TBX3⁺/EPHA4⁺), a SAN-like (SHOX2⁺/NKX2-5⁺/BMPR2⁺) and a TBX18⁺ sinus-venosus subset within a single un-curated annotation label demonstrates that the imputed receptor + TF panel is sufficient to discriminate canonical CCS subtypes from spatial-omics data alone. Three-dimensional spatial positions of the seven subclusters were largely intermingled (Figure 4b), indicating that no clean anatomical regionalisation of these subtypes is detectable at this developmental stage and resolution. Whether this is a real developmental feature of the PCW12 heart or an artefact of MERFISH cell-type label mixing remains to be resolved with orthogonal data.

2.5 Atrial paracrine input is anti-correlated with CCS-TF expression: a niche-driven maturation gradient

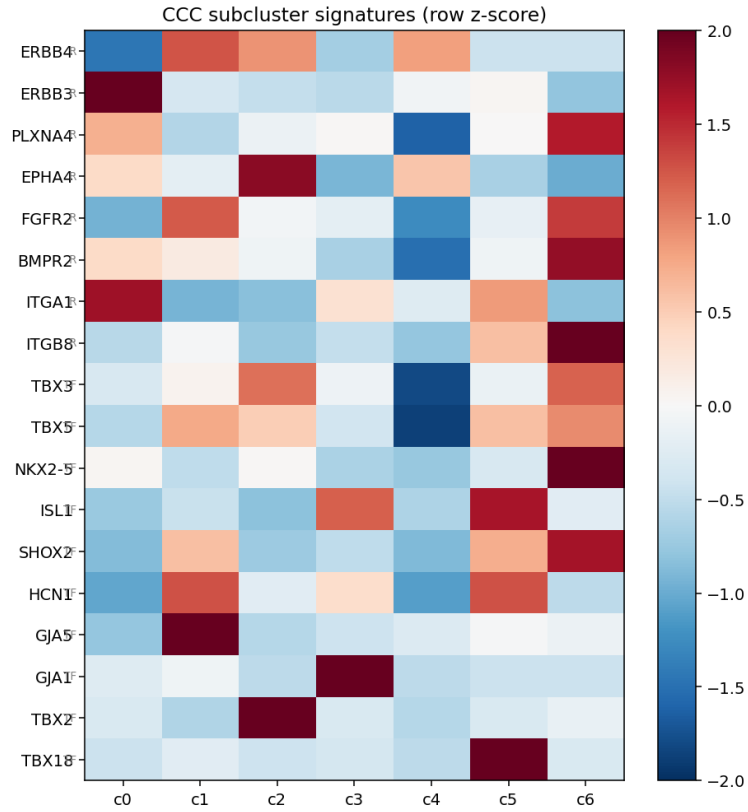
To ask whether ACM-derived signalling input correlates with the molecular state of CCC, we computed a per-CCC *niche score* for each of the two candidate ligands (NRG3, SEMA3A) as the mean ligand expression over the 10 nearest ACM neighbours in 3D. Niche scores were then Spearman-correlated against each of the 10 CCS transcription factors, with multiple-testing correction by the Benjamini–Hochberg procedure [20] over 20 tests (Figure 5a, Table 3).

Strikingly, all significant correlations point in the same direction: **negative**. Nine of ten SEMA3A–TF and four of ten NRG3–TF correlations were significant; in every significant case CCC cells exposed to higher ACM-derived ligand input expressed *lower* levels of CCS transcription factors. The strongest single signal was SEMA3A→NKX2-5 ($\rho = -0.296$, FDR = 5×10^{-6}), followed by NRG3→NKX2-5 ($\rho = -0.254$, FDR = 10^{-4}). Quartile boxplots of NKX2-5 expression as a function of niche-score quartile show a clean, monotonic decrease for both ligands (Figure 5b).



(a) UMAP of 292 CCC, coloured by Leiden cluster.

(b) 3D anatomical position of CCC subclusters.



(c) Mean-expression heatmap of receptors and CCS TFs across the seven subclusters.

Figure 4. CCC heterogeneity reveals canonical CCS subtypes. Leiden clustering on the receptor + CCS-TF panel resolves AV-node-like (c2), SAN-like (c6), TBX18⁺ (c5), pacemaker (c1) and transitional (c3) subsets within the 292-cell pool.

Table 3. Top niche-score $\rightarrow$ CCS-TF correlations (Spearman ρ , BH-FDR over 20 tests).

Ligand niche	TF	ρ	FDR
SEMA3A	NKX2-5	-0.296	5×10^{-6}
SEMA3A	TBX18	-0.236	3×10^{-4}
SEMA3A	TBX5	-0.225	5×10^{-4}
SEMA3A	SHOX2	-0.212	9×10^{-4}
SEMA3A	TBX3	-0.197	2×10^{-3}
NRG3	NKX2-5	-0.254	10^{-4}
NRG3	TBX3	-0.219	6×10^{-4}
NRG3	TBX2	-0.202	2×10^{-3}

Combined with the spatial validation of Section 2.3 — where the same niche-proximal CCC are also ERBB4-high — the data support a coherent two-axis model: **ACM-proximal CCC are ERBB4 $\uparrow$ but CCS-TF $\downarrow$ (responder / less-differentiated), whereas ACM-distal CCC are ERBB4 $\downarrow$ but CCS-TF $\uparrow$ (specified / more mature)**. Atrial NRG/SEMA input thus appears to sustain a less-differentiated CCC pool, while specified pacemaker (c1, c6), AV-node (c2) and TBX18⁺ (c5) cells reside further from the source. This is consistent with neuregulin acting as a survival / proliferation signal during CCS development rather than imposing identity directly [13].

3 Discussion

By coupling entropic optimal-transport imputation [1] with 3D spatial cell-cell interaction analysis [2], we have recovered a genome-scale paracrine interactome of the human fetal heart at PCW12 from a 238-gene MERFISH panel and used it to formulate and test a specific developmental hypothesis: that atrial myocardium acts as a paracrine source whose proximity correlates inversely with conduction-system maturation.

The dominant ACM-derived signal — NRG3 $\rightarrow$ ERBB4 — is consistent with classical genetic studies that established the requirement for NRG/ErbB signalling in cardiac development [11, 12] and specifically for cardiac conduction system formation in mice [13]. Our spatial nearest-neighbour test further demonstrates that this is a *spatially coherent* signal in human tissue, not merely a coincidence of expression: NRG3 in ACM and ERBB4 in adjacent CCC are coupled at $\rho = +0.22$ against a tight permutation null. SEMA3A’s negative spatial coupling with PLXNA4 is consistent with its established role as a repulsive guidance cue and its requirement for normal cardiac rhythm in vivo [14].

Within the CoreConductionCells annotation we resolved at least five biologically interpretable subtypes (Section 2.4), including AV-node-like (TBX2⁺/TBX3⁺ [5]), SAN-like (SHOX2⁺/NKX2-5⁺ [6, 7]) and TBX18⁺ sinus-venosus-derived [8, 9] cells. That a receptor + transcription-factor panel imputed from scRNA-seq onto a target-panel MERFISH measurement is sufficient to recover canonical CCS subtypes argues that the OT-based gene expansion preserves biologically meaningful covariance structure, and not merely average-state information.

Most informatively, the systematic *anti*-correlation of ACM-niche scores with CCS-TF expression (Section 2.5) is the simplest single statement of the data: the CCC cells closest to ACM-derived signalling are the least differentiated cells in the conducting compartment. This is consistent with — though does not prove — a model in which atrial paracrine input sustains a proliferative, less-

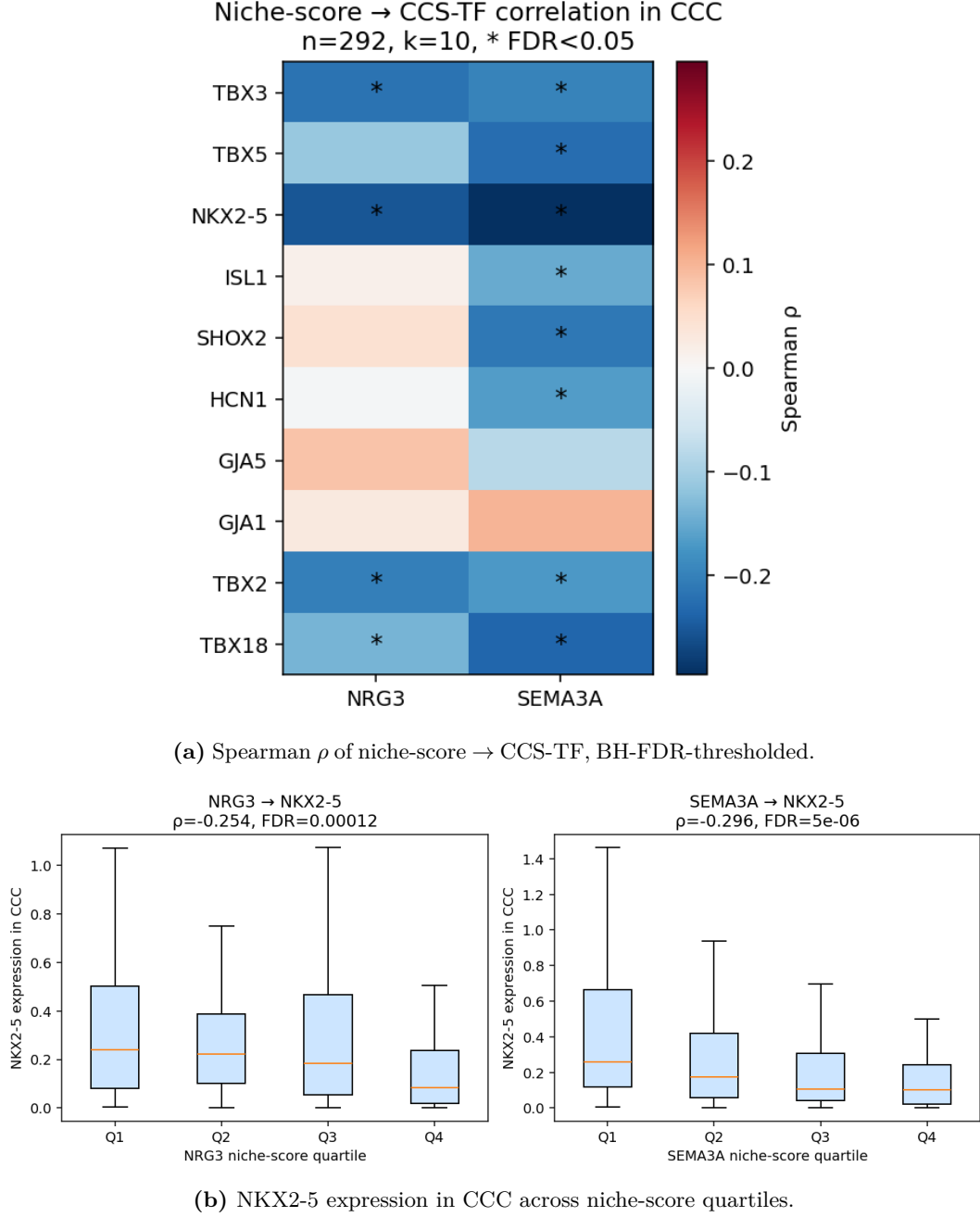


Figure 5. ACM-niche scores are anti-correlated with CCS-TF expression. Negative correlations are systematic across both ligands; NKX2-5 shows a clean monotonic decrease in CCC as the per-cell NRG3/SEMA3A niche score rises.

committed CCC pool while specified pacemaker, AV-node and TBX18⁺ subtypes reside further from the ligand source. The fact that the same CCC cells that are NRG3-niche-high are also ERBB4-high (Section 2.3) strengthens the case that the negative TF correlation is mediated by active receptor engagement rather than mere co-localisation.

Limitations. The analysis has three principal limitations. First, HCN4 and CNTN2, two canonical CCS markers, are not in the 5,187-HVG imputed set; HCN1 was used as a related substitute. A re-run with a CCS-curated supplementary panel would refine cluster identity, particularly for c1. Second, the seven-cluster split is on $n=292$ cells; the biologically smallest clusters (c4 $n=36$, c6 $n=16$) are best treated as candidate identities rather than statistically robust populations. Third, although CCC subclusters were spatially intermingled in 3D, we did not formally annotate atrial vs ventricular regions; orthogonal anatomical labels could test whether the AV-node-like (c2) subset is enriched at the AV junction. Confirmatory work — in particular cross-referencing the c1/c2/c6 subtype-defining transcription factors against matching snATAC peaks and Smart-seq2 trajectory analysis — is a natural next step.

Outlook. The pipeline established here — an imputed spatial atlas, a Spateo CCI loop, spatial nearest-neighbour validation and niche-score regression — is generalisable to other organs and developmental stages. It does not require dedicated assays and works with publicly available image-based spatial panels. We anticipate that analogous gradients structure other developing-organ niches in the human fetus.

4 Methods

Data

The spatial atlas (`full_heart_final_aug2025_update_downsampled_100k.h5ad`) is a 238-gene MERFISH measurement of the human fetal heart at PCW12, downsampled to 10^5 cells, with 3D coordinates in `obsm['X_spateo_update']` and counts log-normalised in `.X`. The single-cell reference (`all_healthy_RoundedPCW12.h5ad`) is a 32,870-cell $\times$ 31,019-gene scRNA-seq dataset with raw counts in `.X` and a hierarchical `celltype` annotation (`coarse_celltype`, `celltype`, `celltype_label`). The disease gene panel comprises 247 unique heart-disease genes from `data/heart_disease_g`

MOSCOT scRNA $\rightarrow$ MERFISH imputation

Following Klein et al. [1], we used the `moscot.problems.MappingProblem` class with `xy_callback="local-pca"` to compute pairwise costs. Marginals were set asymmetric ($\tau_a=1$, $\tau_b=0.9$) to allow some spatial cells to remain unmapped. The mapping was performed in five non-overlapping batches of 2×10^4 spatial cells against the full SC reference; the recovered coupling `pi` was scaled by n_{sp} before imputation. The imputed gene set was the union of the top 5,000 highly variable genes (Seurat flavour, computed on $\log_{10}(1+x)$ -transformed, size-factor-normalised SC counts via `scanpy.pp.highly_variable_genes`) and the 247-gene heart-disease panel, intersected with `adata_sc.var_names`, yielding 5,187 genes (54-gene HVG–disease overlap). Cell-cycle genes were excluded from the OT cost but retained for imputation. Output AnnData (`adata_imputed_hvg_disease.h5ad`; $100,000 \times 5,187$, 4.3 GB) carries `var['is_hvg']` and `var['is_disease']` flags and `obs['mapped_celltype']`/'ma

'mapping_confidence']. Mean and median mapping confidence were 0.49 and 0.45 respectively.

Spatial cell-cell interaction analysis

We used Spateo [2] to compute pairwise spatial cell-cell interactions. After log-normalisation and a 25-nearest-neighbour spatial graph (`spateo.tools.find_neighbors; k=25`), we ran `find_cci_two_group` between every ordered pair of the eight largest mapped cell types (ACM, VCM, FB, Endocardial, MuralCells, NC, CoreConductionCells, TzConductionCells) on a 20,000-cell stratified subsample, with $N_{\text{perm}}=10$ and `filter_lr="outer"`. Top-3 LR rows per ordered pair were written to `results/cci_hvg/all_lr_top3_per_pair.csv`. For the conduction-cell subset (six ordered pairs centred on CCC), we re-ran the loop with $N_{\text{perm}}=50$ to compensate for smaller populations.

Spatial axis validation

For each CCC cell i , we computed the Euclidean nearest neighbour $j(i)$ in 3D within the ACM population and recorded the ligand expression $L_{j(i)}$ (NRG3 or SEMA3A). We then computed the Spearman correlation between $L_{j(i)}$ and the receptor expression R_i (ERBB4 or PLXNA4) across all CCC cells. A permutation null was built by shuffling the ACM-CCC pairing 1,000 times and recomputing the correlation; the empirical p -value is the fraction of permuted $|\rho|$ exceeding the observed $|\rho|$.

CCC subclustering

The 292 CCC cells were extracted, $\log_{10}(1+x)$ -normalised on the 18 panel features (8 receptors, 10 CCS-TFs), PCA'd to 10 components, `scanpy.pp.neighbors` with $k=15$, and clustered with the Leiden algorithm [19] at resolution 0.8. Cluster markers were computed across the full 5,187-gene set with `scanpy.tl.rank_genes_groups` (Wilcoxon, BH-corrected).

Niche-score / TF correlation

For each CCC cell i , the niche score for ligand ℓ was defined as the mean of ℓ over the $k=10$ nearest ACM neighbours of i . We computed the Spearman correlation between each (ligand, CCS-TF) pair across all CCC and applied Benjamini-Hochberg correction [20] across the 20 tests. Quartile boxplots were generated by binning CCC cells by quartile of niche score and plotting the distribution of TF expression.

Software

Analyses were performed in Python 3.11 with `scanpy` 1.10 [18], `moscot` 0.5.0 [1], `spateo` 1.1 [2], `leidenalg` [19], `scipy`, `numpy`, `pandas` and `matplotlib`. All scripts and intermediate artefacts are reproducible from `PCW12_analysis/scripts/{20-28}*`.

Data and code availability

All analysis scripts, log files, intermediate AnnData artefacts, results CSVs and figure source files are organised under `PCW12_analysis/` in the project workspace. The paper and its LaTeX source live under `PCW12_analysis/paper/`.

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